Low Voltage 16-bit D-type Flip-flop With 5 V Tolerant Inputs and Outputs

74LCX16374

The LCX16374 contains sixteen non-inverting D-type flip-flops with 3-STATE outputs and is intended for bus oriented applications. The device is byte controlled. A buffered clock (CP) and Output Enable ($\overline{OE}$) are common to each byte and can be shorted together for full 16-bit operation.

The LCX16374 is designed for low voltage (2.5 V or 3.3 V) V_{CC} applications with capability of interfacing to a 5 V signal environment.

The LCX16374 is fabricated with an advanced CMOS technology to achieve high speed operation while maintaining CMOS low power dissipation.

Features

- 5 V Tolerant Inputs and Outputs
- 2.3 V to 3.6 V V_{CC} Specifications Provided
- 6.2 ns t_{PD} Max (V_{CC} = 3.3 V), 20 μA I_{CC} Max
- Power-Down High-Impedance Inputs and Outputs
- Supports Live Insertion and Withdrawal (Note 1)
- ±24 mA Output Drive (V_{CC} = 3.0 V)
- Uses Proprietary Noise/EMI Reduction Circuitry
- Latch-Up Performance Exceeds 500 mA
- ESD Performance:
  - Human Body Model: 2000 V
  - Machine Model: 200 V
- Also Packaged in Plastic Fine-Pitch Ball Grid Array (FBGA)

1. To ensure the high-impedance state during power up or down, $\overline{OE}$ should be tied to V_{CC} through a pull-up resistor: the minimum value or the resistor is determined by the current-sourcing capability of the driver.

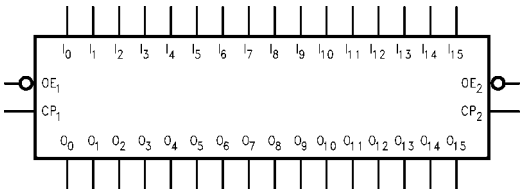
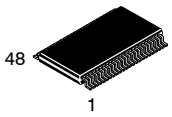
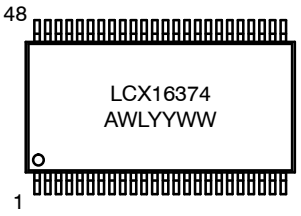


Figure 1. Logic Symbol



TSSOP48 12.5x6.1
CASE 1201

MARKING DIAGRAM



A	= Assembly Location
WL	= Wafer Lot
YY	= Year
WW	= Work Week

ORDERING INFORMATION

See detailed ordering and shipping information on page 8 of this data sheet.

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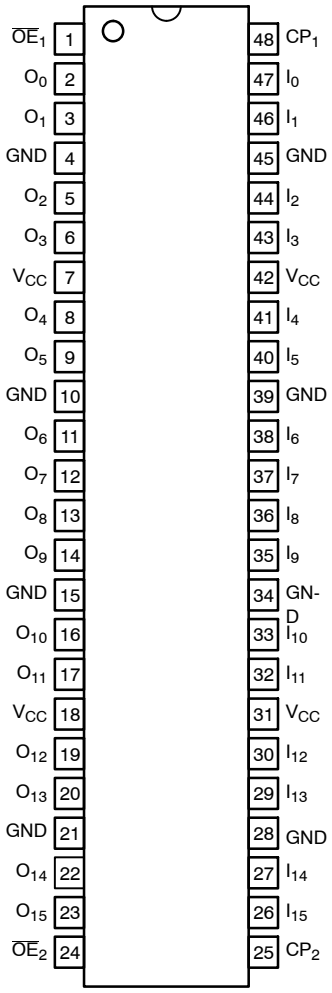


Figure 2. Pin Assignment for SSOP and TSSOP

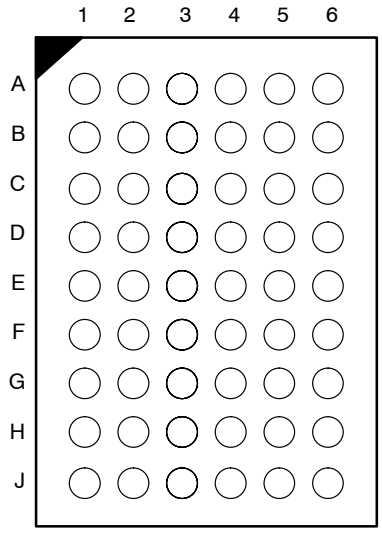


Figure 3. Pin Assignment for FBGA

PIN NAMES

Pins	Function
$\overline{OE}_n$	Output Enable Inputs (Active Low)
CP_n	Clock Pulse Inputs
I_0–I_{15}	Inputs
O_0–O_{15}	Outputs
NC	No Connect

TRUTH TABLE

Inputs			Outputs
CP_1	$\overline{OE}_1$	I_0:I_7	O_0:O_7
↑	L	H	H
↑	L	L	L
L	L	X	O_0
X	H	X	Z

TRUTH TABLE

Inputs			Outputs
CP_2	$\overline{OE}_2$	I_8:I_{15}	O_8:O_{15}
↑	L	H	H
↑	L	L	L
L	L	X	O_0
X	H	X	Z

H = High Voltage Level
L = Low Voltage Level
X = Immaterial
Z = High Impedance
O_0 = Previous O_0 before HIGH-to-LOW of CP

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FBGA PIN ASSIGNMENTS

	1	2	3	4	5	6
A	O_0	NC	$\overline{OE}_1$	CP_1	NC	I_0
B	O_2	O_1	NC	NC	I_1	I_2
C	O_4	O_3	V_{CC}	V_{CC}	I_3	I_4
D	O_6	O_5	GND	GND	I_5	I_6
E	O_8	O_7	GND	GND	I_7	I_8
F	O_{10}	O_9	GND	GND	I_9	I_{10}
G	O_{12}	O_{11}	V_{CC}	V_{CC}	I_{11}	I_{12}
H	O_{14}	O_{13}	NC	NC	I_{13}	I_{14}
J	O_{15}	NC	$\overline{OE}_2$	CP_2	NC	I_{15}

Functional Description

The LCX16374 consists of sixteen edge-triggered flip-flops with individual D-type inputs and 3-STATE true outputs. The device is byte controlled, with each byte functioning identically but independently of the other. The control pins can be shorted together to obtain full 16-bit operation. Each byte has a buffered clock and buffered Output Enable common to all flip-flops within that byte. The description that follows applies to each byte. Each flip-flop stores the state of its individual D input that meets the setup and hold time requirements on the LOW-to-HIGH Clock (CP_n) transition. When Output Enable ($\overline{OE}_n$) is LOW, the contents of the flip-flops are available at the outputs. When $\overline{OE}_n$ is HIGH, the outputs enter the high-impedance state. Operation of the $\overline{OE}_n$ input does not affect the state of the flip-flops.

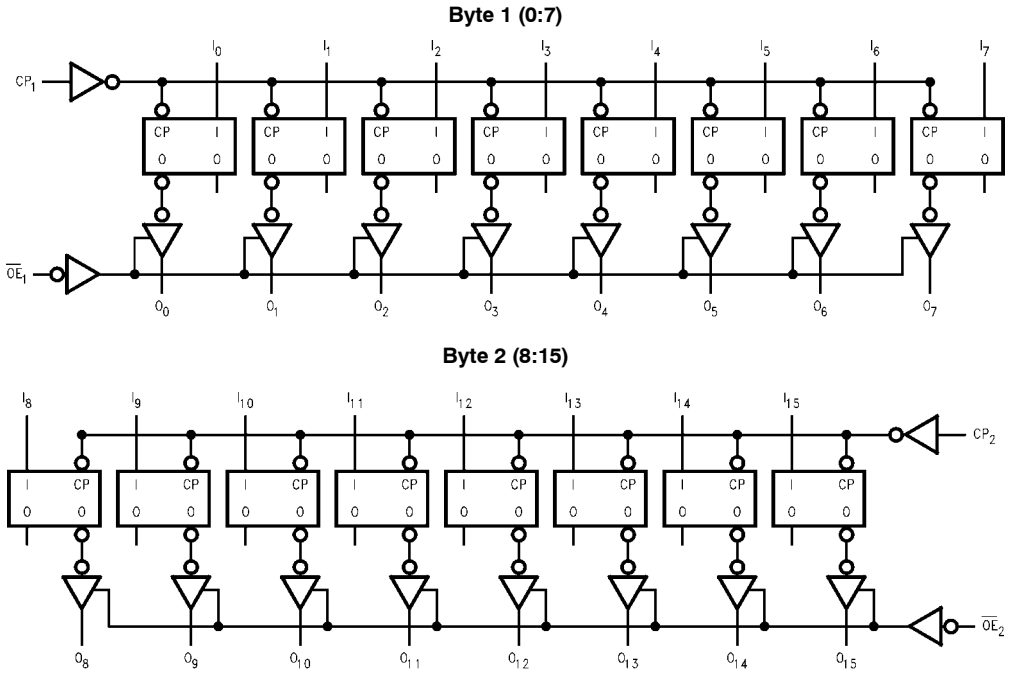


Figure 4. Logic Diagram

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MAXIMUM RATINGS

Symbol	Parameter	Conditions	Value	Unit
V_{CC}	Supply Voltage		−0.5 to +6.5	V
V_I	DC Input Voltage		−0.5 to +6.5	V
V_O	DC Output Voltage	3-STATE	−0.5 to +6.5	V
		Output in HIGH or LOW State (Note 1)	−0.5 to VCC + 0.5	
V_O	DC Output Voltage	Power-Down Mode (V_{CC} = 0 V)	−0.5 to +6.5	V
I_{IK}	DC Input Diode Current	V_I < GND	−50	mA
I_{OK}	DC Output Diode Current	V_O < GND	−50	mA
		V_O > V_{CC}	+50	
I_O	DC Output Source/Sink Current		±50	mA
I_{CC}	DC Supply Current per Supply Pin		±100	mA
I_{GND}	DC Ground Current per Ground Pin		±100	mA
T_{STG}	Storage Temperature		−65 to +150	°C
T_L	Lead Temperature, 1 mm from Case for 10 Seconds		260	°C
T_J	Junction Temperature under Bias		+150	°C
MSL	Moisture Sensitivity		Level 1	
F_R	Flammability Rating	Oxygen Index: 28 to 34	UL 94 V−0 @ 0.125 in	
V_{ESD}	ESD Withstand Voltage (Note 2)	Human Body Model	2000	V
		Charged Device Model	N/A	

Stresses exceeding those listed in the Maximum Ratings table may damage the device. If any of these limits are exceeded, device functionality should not be assumed, damage may occur and reliability may be affected.

1. I_O absolute maximum rating must be observed.
2. HBM tested to EIA / JESD22−A114−A. CDM tested to JESD22−C101−A. JEDEC recommends that ESD qualification to EIA/JESD22−A115A (Machine Model) be discontinued.

RECOMMENDED OPERATING CONDITIONS

Symbol	Parameter		Min	Typ	Max	Unit
V_{CC}	Supply Voltage	Operating	1.65	2.5, 3.3	5.5	V
		Data Retention Only	1.5	2.5, 3.3	5.5	
V_I	Digital Input Voltage		0	–	5.5	V
V_O	Output Voltage	Active Mode (High or Low State)	0	–	V_{CC}	V
		Tri-State Mode	0	–	5.5	
		Power Down Mode (V_{CC} = 0 V)	0	–	5.5	
T_A	Operating Free-Air Temperature		−40	–	+125	°C
t_r, t_f	Input Rise or Fall Rate	V_{CC} = 1.65 V to 1.95 V	0	–	20	nS/V
		V_{CC} = 2.3 V to 2.7 V	0	–	20	
		V_{IN} from 0.8 V to 2.0 V, V_{CC} = 3.0 V	0	–	10	
		V_{CC} = 4.5 V to 5.5 V	0	–	5	

Functional operation above the stresses listed in the Recommended Operating Ranges is not implied. Extended exposure to stresses beyond the Recommended Operating Ranges limits may affect device reliability.

3. Unused inputs must always be tied to an appropriate logic voltage level (e.g., either GND or V_{CC}). Unused outputs must be left open.

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DC ELECTRICAL CHARACTERISTICS

Symbol	Parameter	Conditions	V_{CC} (V)	T_A = −40 °C to +85 °C		T_A = −40 °C to +125 °C		Unit
				Min	Max	Min	Max	
V_{IH}	HIGH Level Input Voltage		1.65 – 1.95	0.65 x V_{CC}	–	0.65 x V_{CC}	–	V
			2.3 – 2.7	1.7	–	1.7	–	
			3.0 – 3.6	2.0	–	2.0	–	
			4.5 – 5.5	0.70 x V_{CC}	–	0.70 x V_{CC}	–	
V_{IL}	LOW Level Input Voltage		1.65 – 1.95	–	0.35 x V_{CC}	–	0.35 x V_{CC}	V
			2.3 – 2.7	–	0.7	–	0.7	
			3.0 – 3.6	–	0.8	–	0.8	
			4.5 – 5.5	–	0.30 x V_{CC}	–	0.30 x V_{CC}	
V_{OH}	High-Level Output Voltage	V_I = V_{IH} or V_{IL}						V
		I_{OH} = −100 μA	1.65 to 5.5	V_{CC} – 0.1	–	V_{CC} – 0.1	–	
		I_{OH} = −4 mA	1.65	1.29	–	1.29	–	
		I_{OH} = −8 mA	2.3	1.8	–	1.8	–	
		I_{OH} = −12 mA	2.7	2.2	–	2.2	–	
		I_{OH} = −16 mA	3.0	2.4	–	2.4	–	
		I_{OH} = −24 mA	3.0	2.2	–	2.2	–	
		I_{OH} = −32 mA	4.5	3.7	–	3.7	–	
V_{OL}	Low-Level Output Voltage	V_I = V_{IH} or V_{IL}						V
		I_{OL} = 100 μA	1.65 to 5.5	–	0.1	–	0.1	
		I_{OL} = 4 mA	1.65	–	0.24	–	0.24	
		I_{OL} = 8 mA	2.3	–	0.3	–	0.3	
		I_{OL} = 12 mA	2.7	–	0.4	–	0.4	
		I_{OL} = 16 mA	3.0	–	0.4	–	0.4	
		I_{OL} = 24 mA	3.0	–	0.55	–	0.55	
		I_{OL} = 32 mA	4.5	–	0.6	–	0.6	
I_I	Input Leakage Current	V_I = 0 to 5.5 V	3.6	–	±5.0	–	±5.0	μA
I_{OFF}	Power Off Leakage Current	V_I = 5.5 V or V_O = 5.5 V	0	–	10	–	10	μA
I_{CC}	Quiescent Supply Current	V_I = 5.5 V or GND	3.6	–	10	–	10	μA
ΔI_{CC}	Increase in I_{CC} per Input	V_{IH} = V_{CC} – 0.6 V	2.3 to 3.6	–	500	–	500	μA

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AC ELECTRICAL CHARACTERISTICS

Symbol	Parameter	T_A = −40 °C to +125 °C										Unit
		V_{CC} = 5 V ± 0.5 V, C_L = 50 pF		V_{CC} = 3.3 V ± 0.3 V, C_L = 50 pF		V_{CC} = 2.7 V, C_L = 50 pF		V_{CC} = 2.5 V ± 0.2 V, C_L = 30 pF		V_{CC} = 1.8 V ± 0.15 V, C_L = 30 pF		
		Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
f_{MAX}				170								MHz
t_{PHL}, t_{PLH}	Propagation Delay, CP to O_n		6.2 6.2	1.5 1.5	6.2 6.2	1.5 1.5	6.5 6.5	1.5 1.5	7.4 7.4		12.4 12.4	ns
t_{PZL}, t_{PZH}	Output Enable Time		6.1 6.1	1.5 1.5	6.1 6.1	1.5 1.5	6.3 6.3	1.5 1.5	7.9 7.9		12.4 12.4	ns
t_{PLZ}, t_{PHZ}	Output Disable Time		6.0 6.0	1.5 1.5	6.0 6.0	1.5 1.5	6.2 6.2	1.5 1.5	7.2 7.2		9.1 9.1	ns
t_S	Setup Time	2.5		2.5		2.5		3.0		3.0		ns
t_H	Hold Time	1.5		1.5		1.5		2.0		2.5		ns
t_W	Pulse Time	3.0		3.0		3.0		3.5		5.0		ns
t_{OSHL}, t_{OSLH}	Output to Output Skew (Note 4)		1.0 1.0		1.0 1.0							ns

Product parametric performance is indicated in the Electrical Characteristics for the listed test conditions, unless otherwise noted. Product performance may not be indicated by the Electrical Characteristics if operated under different conditions.

4. Skew is defined as the absolute value of the difference between the actual propagation delay for any two separate outputs of the same device. The specification applies to any outputs switching in the same direction, either HIGH-to-LOW (t_{OSHL}) or LOW-to-HIGH (t_{OSLH}).

DYNAMIC SWITCHING CHARACTERISTICS

Symbol	Characteristic	Condition	T_A = +25 °C	Unit
			Typ	
V_{OLP}	Quiet Output Dynamic Peak V_{OL}	C_L = 50 pF, V_{IH} = 3.3 V, V_{IL} = 0 V C_L = 30 pF, V_{IH} = 2.5 V, V_{IL} = 0 V	0.8 0.6	V
V_{OLV}	Quiet Output Dynamic Valley V_{OL}	C_L = 50 pF, V_{IH} = 3.3 V, V_{IL} = 0 V C_L = 30 pF, V_{IH} = 2.5 V, V_{IL} = 0 V	−0.8 −0.6	V

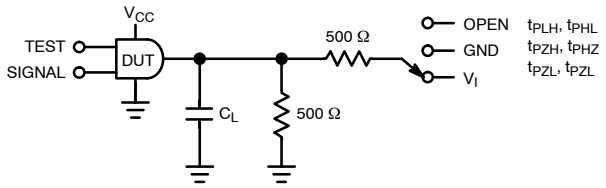
5. Number of outputs defined as “n”. Measured with “n-1” outputs switching from HIGH-to-LOW or LOW-to-HIGH. The remaining output is measured in the LOW state.

CAPACITIVE CHARACTERISTICS

Symbol	Parameter	Condition	Typ	Unit
C_{IN}	Input Capacitance	V_{CC} = 3.3 V, V_I = 0 V or V_{CC}	7	pF
C_{OUT}	Output Capacitance	V_{CC} = 3.3 V, V_I = 0 V or V_{CC}	8	pF
C_{PD}	Power Dissipation Capacitance	V_{CC} = 3.3 V, V_I = 0 V or V_{CC}, f = 10 MHz	20	pF

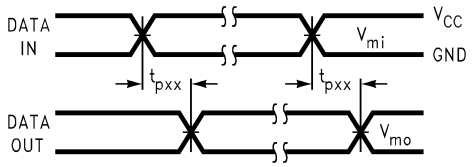
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AC LOADING AND WAVEFORMS (GENERIC FOR LCX FAMILY)

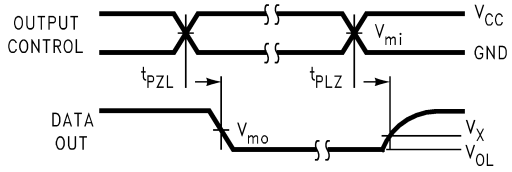


Test	Switch Position
t_{PLH} / t_{PHL}	Open
t_{PLZ} / t_{PZL}	6 V at V_{CC} = 3.3 ± 0.3 V and 2.7 V V_{CC} × 2 at V_{CC} = 2.5 ± 0.2 V
t_{PHZ} / t_{PZH}	GND

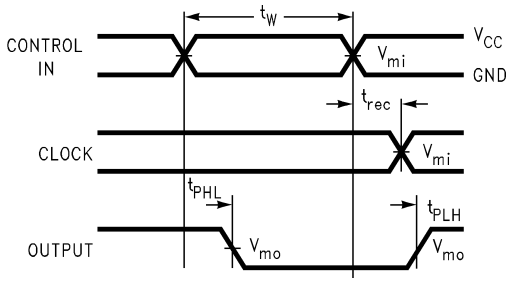
Figure 5. AC Test Circuit (C_L Includes Probe and Jig Capacitance)



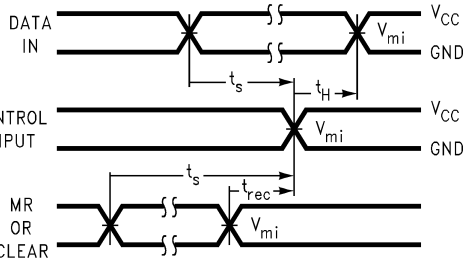
Waveform for Inverting and Non-Inverting Functions



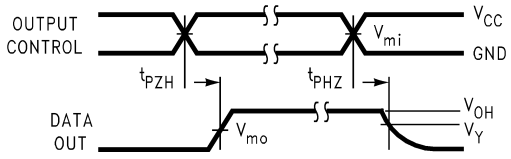
3-STATE Output High Enable and Disable Times for Logic



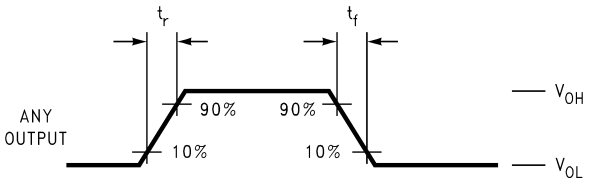
Propagation Delay. Pulse Width and t_{rec} Waveforms



Setup Time, Hold Time and Recovery Time for Logic



3-STATE Output Low Enable and Disable Times for Logic



t_{rise} and t_{fall}

Symbol	VCC		
	3.3 V ± 0.3 V	2.7 V	2.5 V ± 0.2 V
V_{mi}	1.5 V	1.5 V	V_{CC} / 2
V_{mo}	1.5 V	1.5 V	V_{CC} / 2
V_x	V_{OL} + 0.3 V	V_{OL} + 0.3 V	V_{OL} + 0.15 V
V_y	V_{OH} − 0.3 V	V_{OH} − 0.3 V	V_{OH} − 0.15 V

Figure 6. Waveforms (Input Characteristics; f = 1 MHz, t_r = t_f = 3 ns)

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SCHEMATIC DIAGRAM (Generic for LCX Family)

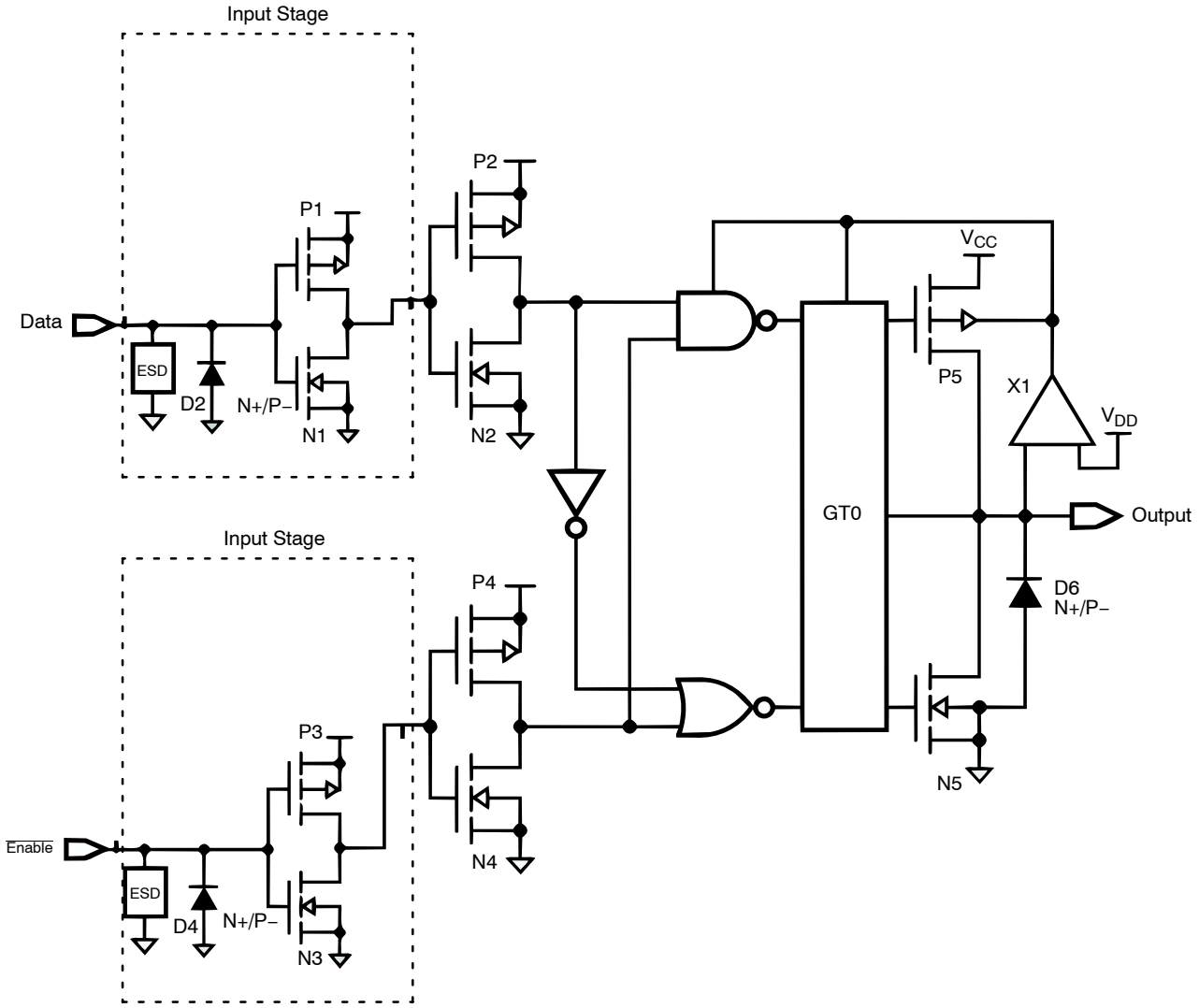


Figure 7. Schematic Diagram (Generic for LCX Family)

ORDERING INFORMATION

Product Number	Marking	Package	Shipping†
74LCX16374MTDX	LCX16374	TSSOP-48	1000 / Tape & Reel

DISCONTINUED (Note 6)

74LCX16374G			1000 / Tape & Reel
74LCX16374MEA			1000 / Tape & Reel
74LCX16374MTD	$Y&Z&2&K LCX16374	TSSOP-48	3800 Units / Tube

† For information on tape and reel specifications, including part orientation and tape sizes, please refer to our Tape and Reel Packaging Specifications Brochure, BRD8011/D.

6. **DISCONTINUED:** These devices are not available. Please contact your **onsemi** representative for information. The most current information on these devices may be available on www.onsemi.com.

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REVISION HISTORY

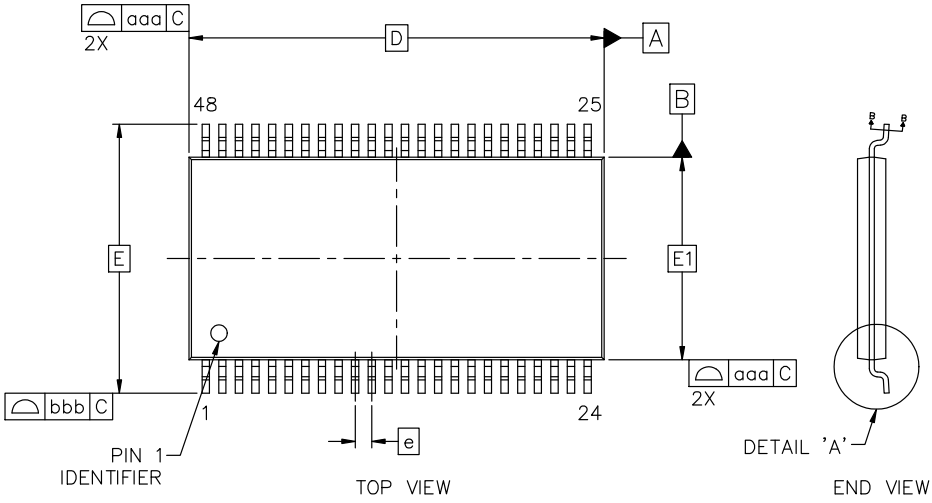
Revision	Description of Changes	Date
2	Revised specification table. Document converted to **onsemi** format. 74LCX16374MTD OPN Marked as Discontinued.	8/20/2026
3	Changed package outline from Case 948BQ to Case 1201.	9/22/2026

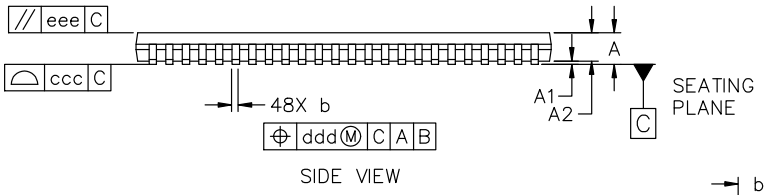
This document has undergone updates prior to the inclusion of this revision history table. The changes tracked here only reflect updates made on the noted approval dates.

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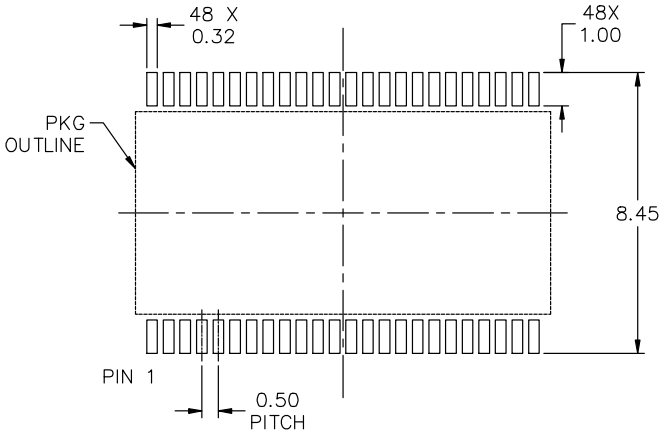
PACKAGE DIMENSIONS

TSSOP−48 12.50x6.10x0.85, 0.50P
CASE 1201−01
ISSUE C



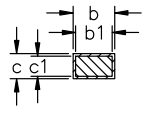


MILLIMETERS			
DIM	MIN	NOM	MAX
A	− − −	− − −	1.10
A1	0.05	0.10	0.15
A2	0.75	0.85	0.95
b	0.17	0.22	0.27
b1	0.17	0.20	0.23
c	0.09	0.15	0.20
c1	0.09	0.12	0.16
D	12.50 BSC		
E	8.10 BSC		
E1	6.10 BSC		
e	0.50 BSC		
L	0.50	0.60	0.70
L1	1.00 REF		
L2	0.25 BSC		
θ	0°	4°	8°
TOLERANCE FORM & POSITION			
aaa	0.10		
bbb	0.20		
ccc	0.10		
ddd	0.13		
eee	0.10		

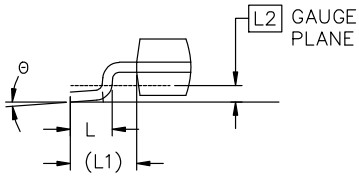


RECOMMENDED MOUNTING FOOTPRINT

*For additional information on our Pb−Free strategy and soldering details, please download the onsemi Soldering and Mounting Techniques Reference Manual, SOLDERRM/D.



DETAIL 'B−B'
SCALE 5:1



DETAIL 'A'
SCALE 2:1

NOTES:
1. DIMENSIONING AND TOLERANCING AS PER ASME Y14.5M, 2018.
2. CONTROLLING DIMENSION: MILLIMETERS.
3. DIMENSIONS D AND E1 DO NOT INCLUDE MOLD FLASH, PROTRUSIONS OR GATE BURRS. MOLD FLASH OR GATE BURRS SHALL NOT EXCEED 0.15 PER SIDE.
4. DIMENSION b DOES NOT INCLUDE DAMBAR PROTRUSION. ALLOWABLE DAMBAR PROTRUSION SHALL BE 0.08 TOTAL IN EXCESS OF THE b DIMENSION AT MAXIMUM MATERIAL CONDITION.
5. TERMINAL NUMBERS ARE SHOWN FOR REFERENCE ONLY.

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